



Test 2 FSPM 0024 Calculus 20212022-3

CALCULUS - foundation (Universiti Teknologi Malaysia)



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Foundation Programme
Universiti Teknologi Malaysia
Session 2021/2022 Semester 3

Test 2 (20%)

Time: 90 minutes

FSPM 0024 Calculus

Answer all questions. Show all your work clearly. Any similarity with answers given by apps/chegg or any websites will be punished accordingly. List of formula is available on the last page.

1. Given $\int_0^5 g(x) dx = 2$, $\int_0^3 g(x) dx = -4$, $\int_3^5 [5g(x) - 4h(x)] dx = 50$.

i. Find $\int_3^5 h(x) dx$.

(3 marks)

ii. Hence, evaluate $\int_3^5 \left[\frac{3}{2}x^3 - 11h(x) \right] dx$.

(3 marks)

2. By taking $u^2 = 3x + 1$, solve the integral

$$\int \frac{2x + 4}{\sqrt{3x + 1}} dx.$$

(5 marks)

3. Evaluate $\int e^{2x} \sin 4x dx$ by using tabular method.

(4 marks)

4. By using the substitution $u = e^x$, show that

$$\int_{-1}^1 \frac{e^x}{e^x + 1} dx = 1.$$

(5 marks)

5. Given

$$f(x) = \frac{12x^2 + x + 3}{(6x + 1)(2x^2 + 1)}.$$

i. Find the partial fraction of $f(x)$.

(3 marks)

ii. Evaluate $\int_0^2 f(x) dx$. Give your answer in terms of natural logarithm.

(5 marks)

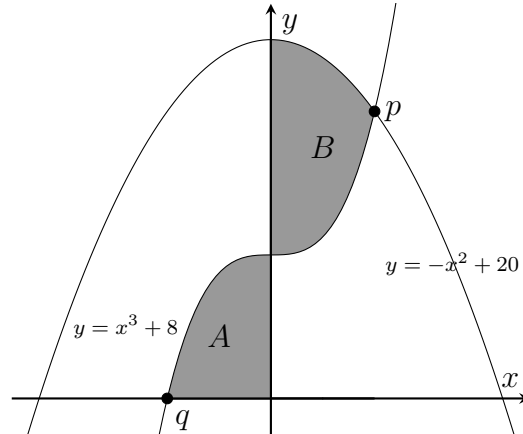


Figure 1: Figure for Question 6

6. Figure 1 shows the regions A and B that are bounded by the curves $y = x^3 + 8$, $y = -x^2 + 20$, x -axis and y -axis. Point p is the intersection point between the two curves while point q is located on the x -axis.

i. State the x -coordinates for p and q .

(2 marks)

ii. Find the total area of region A and B .

(5 marks)

iii. Find the volume of solid revolution of region A formed by revolving the region 360° about the y -axis.

(5 marks)

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$ • $1 + \tan^2 x = \sec^2 x$ • $\cot^2 x + 1 = \operatorname{cosec}^2 x$ • $\sin 2x = 2 \sin x \cos x$ • $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ • $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ • $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ • $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ • $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ • $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ • $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Differentiations	Integrations
$\frac{d}{dx}[k] = 0, k \text{ constant}$	$\int k dx = kx + C, k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
$\frac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\int \frac{dx}{x} = \ln x + C$
$\frac{d}{dx}[\cos x] = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x dx = -\cot x + C$
$\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$